

Title: Automation Testing Plan using Selenium with Robot Framework for Enterprise Dashboard Application

1. Overview

This plan describes how to implement a functional UI test automation framework using Selenium with Robot Framework for your enterprise application.

Technology Stack Recap

Frontend: Next.js (TypeScript, App Router)

State Management: Redux Toolkit

Backend: Java Spring Boot

Database: Snowflake

Authentication: SSO + Role-Based Access Control (RBAC)

Unit Testing: Jest (UI), JUnit (Java API)

CI/CD: GitHub Actions + AWS

Robot Framework provides keyword-driven testing, making it readable and beginner-friendly while being powerful for complex workflows.

2. Why Robot Framework?

Easy to write and read test cases

Good integration with Selenium Library

Works well for acceptance and regression testing

Supports tags, variables, and reusable test resources

Integrates easily with GitHub Actions

3. Tools and Libraries

Robot Framework (core framework)

SeleniumLibrary (for UI automation)

Browser Drivers (Chrome, Firefox)

Python (interpreter, environment)

Allure Robot Framework (for reporting)

4. Directory Structure

```
robot-tests/
├── resources/           # Page-specific keywords
│   ├── LoginResource.robot
│   ├── DashboardResource.robot
│   └── WorkflowResource.robot
├── testcases/          # Test suite files
│   ├── LoginTests.robot
│   ├── DashboardTests.robot
│   └── WorkflowTests.robot
├── variables/          # Global vars like credentials
│   └── global_variables.robot
├── results/            # Test execution results
├── requirements.txt     # Python dependencies
└── robot.yaml          # GitHub Actions workflow
```

5. Setup Instructions

Install Prerequisites

```
pip install robotframework
pip install robotframework-seleniumlibrary
pip install webdriver-manager
```

Start Sample Script

*** Settings ***

```
Library      SeleniumLibrary
Variables    ../variables/global_variables.robot
Resource     ../resources/LoginResource.robot
```

*** Test Cases ***

Login With Valid SSO User

```
[Tags]  smoke  login
Open Browser  ${APP_URL}  ${BROWSER}
Login As SSO User  ${USERNAME}  ${PASSWORD}
Page Should Contain Element  xpath=//div[@class='dashboard']
Close Browser
```

6. Sample Page Keyword File

*** Keywords ***

Login As SSO User

```
Input Text  id=username  ${USERNAME}
Input Text  id=password  ${PASSWORD}
Click Button  id=login-button
```

7. Sample Variables File

*** Variables ***

```
${APP_URL}  https://your-application.com
${BROWSER}  chrome
${USERNAME}  testuser
${PASSWORD}  testpassword
```

8. Suggested Test Cases

Login and SSO

Valid SSO login

Invalid credentials

Role-based menu visibility

Dashboard

Validate chart rendering

Table filtering, sorting

Pagination navigation

Workflow Wizard

Complete multi-step wizard

Validation errors on required fields

Approve and reject workflows

9. CI/CD with GitHub Actions

```
# .github/workflows/robot-tests.yml
name: Run Robot Framework Tests
```

```
on: [push]
```

```
jobs:
```

```
  robot-tests:
```

```
    runs-on: ubuntu-latest
```

steps:

- uses: actions/checkout@v3
- name: Set up Python
 - uses: actions/setup-python@v4
 - with:
 - python-version: '3.11'
- name: Install dependencies
 - run: |
 - pip install -r requirements.txt
- name: Run Robot Framework Tests
 - run: |
 - robot -d results testcases/

10. Team Roles and Responsibilities

Role	Task Description
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QA Lead	Framework architecture and governance
Automation Testers	Create resource files and test cases
Dev Team	Add locators and test hooks in UI
DevOps	Integrate test run with CI/CD pipelines

11. Next Steps

1. Create resource and variable files for each app section
2. Automate login and RBAC checks first
3. Move to dashboards and workflows
4. Parameterize for different user roles

5. Integrate with GitHub Actions and schedule nightly runs

Let me know if you want a working GitHub project or help writing tests for specific dashboards or workflows.

I've created a detailed document outlining a complete automation test plan using Selenium with Robot Framework, tailored to your tech stack, SSO, and RBAC requirements. You can find it in the Selenium Robot Framework Plan document.

Let me know if you need:

Sample test cases for your dashboards

A GitHub starter project

CI/CD integration help with GitHub Actions

I'm happy to walk you through the setup step-by-step.